Aim: Implementation of nesC Programming Model

Create a basic nesC application using modules and configurations to understand component wiring and interface binding.

CODE:

>>BlinkC.nc

```
#include "Timer.h"

module BlinkC {

   // Interfaces used by this module
   uses interface Boot;
   uses interface Leds;
   uses interface Timer<TMilli> as Timer0;
}

implementation {

   event void Boot.booted() {

      // Command: start timer
      call Timer0.startPeriodic(500);
   }

   event void Timer0.fired() {

      // Command: toggle LED
      call Leds.led0Toggle();
   }
}
```

>>BlinkAppC.nc

```
configuration BlinkAppC {
}

implementation {

   // 1. Declare the components
   components MainC;
   components LedsC;
   components new TimerMilliC() as Timer0;
   components BlinkC as App;
```

Jeetesh Shirganoor

```
    // 2. Wire the interfaces (Interface Binding)
    App.Boot -> MainC.Boot;
    App.Leds -> LedsC;
    App.Timer0 -> Timer0;
}
```

>>sim.py

```
import sys
from TOSSIM import *

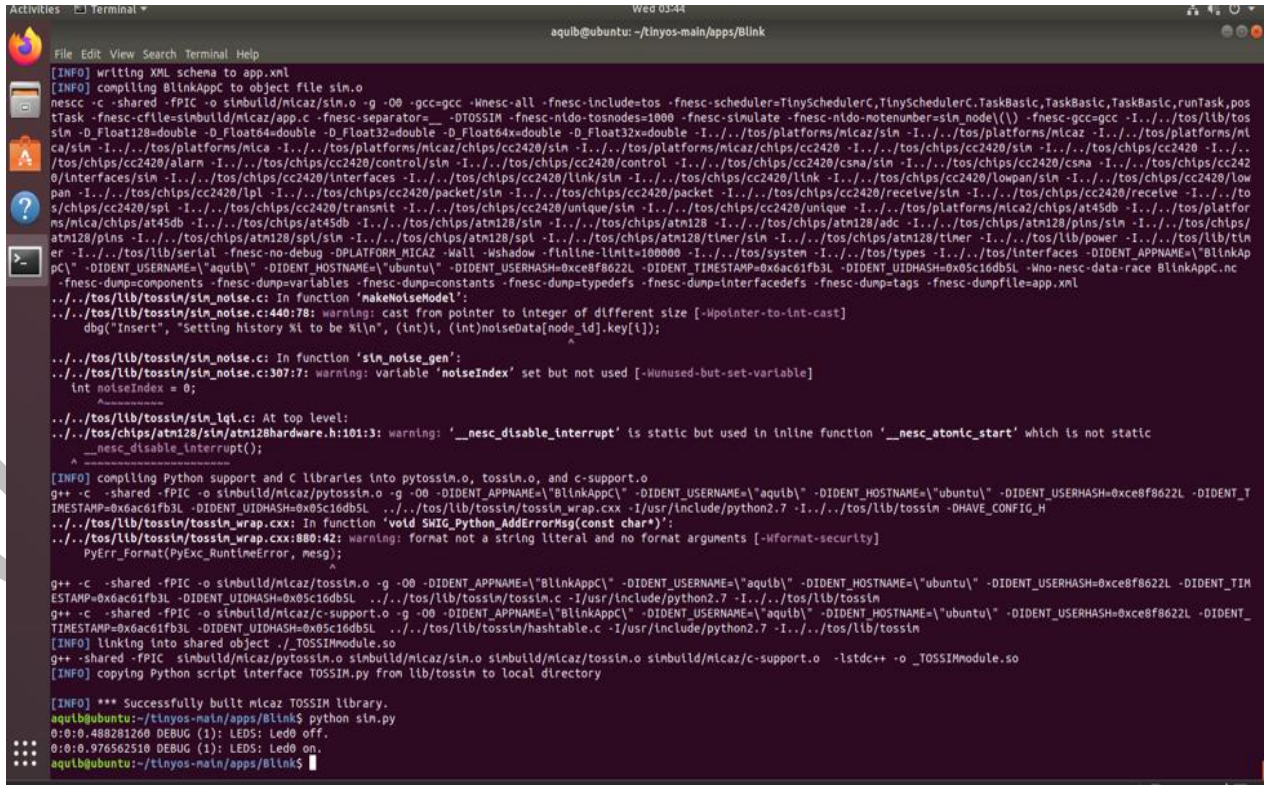
t = Tossim([])

t.addChannel("LedsC", sys.stdout)

# Boot mote 1
m = t.getNode(1)
m.bootAtTime(0)

# Run 20 events
for i in range(20):
    t.runNextEvent()
```

OUTPUT :



Jeetesh Shirganoor